



STEM SMART Single Maths 46 - Pure (non-calculus) Revision >

Arithmetic Series 3i

Subject & topics: Maths **Stage & difficulty:** A Level P2

A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 8 \quad \text{and} \quad u_{n+1} = u_n + 3.$$

Part A

u_5

Find u_5 .

The following symbols may be useful: `u_5`

Part B

Terms in the sequence

The n^{th} term of the sequence can be written in the form $u_n = pn + q$. State the values of p and q .

$p =$

$q =$

Part C

Type of sequence

What type of sequence is it?

- ☐ Arithmetic progression
- ☐ Geometric progression
- ☐ Periodic sequence

Part D

Value of N

Find the value of N such that $\sum_{n=1}^{2N} u_n - \sum_{n=1}^N u_n = 1256$.

The following symbols may be useful: $\mathbb{N}$

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Geometric Series 5i

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Value of k

The first term of a geometric progression is 50 and the common ratio is 0.8. Use logarithms to find the smallest value of k such that the k^{th} term is less than 0.15.

The following symbols may be useful: k

Part B

Common ratio and first term

In a different geometric progression, the second term is -3 and the sum to infinity is 4. Find the common ratio. Hence, find the first term.

Enter your answers below. If a value is not a whole number, enter the value as a decimal.

common ratio =

first term =

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Binomial: All Rational n 1i

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Partial Fractions

Given that $\frac{3x + 4}{(1 + x)(2 + x)^2} \equiv \frac{A}{1 + x} + \frac{B}{2 + x} + \frac{C}{(2 + x)^2}$, find A , B , and C .

If a value is not a whole number, enter the value as a decimal.

$A =$

$B =$

$C =$

Part B

Expand

Hence or otherwise expand $\frac{3x + 4}{(1 + x)(2 + x)^2}$ in ascending powers of x , up to and including the term in x^2 .

The following symbols may be useful: x

Part C

Values of x

State the set of values of x for which the expansion in the above part is valid.

Construct your answer from the items below.

Items:

x

$<$

$\leq$

$>$

$\geq$

$< x <$

$\leq x \leq$

$> x \text{ or } x >$

$\geq x \text{ or } x \geq$

-4

-2

$-\frac{3}{2}$

-1

$-\frac{1}{2}$

0

$\frac{1}{2}$

1

$\frac{3}{2}$

2

4

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Circles 4i

Subject & topics: Maths **Stage & difficulty:** A Level P1

A circle with centre C has the equation $x^2 + y^2 - 10x + 4y + 4 = 0$.

Part A

Find the coordinates of C

By completing the square for x and y find the coordinates of C.

Enter the x and y coordinates below. If a value is not a whole number, enter the value as a decimal.

(,)

Part B

Find radius

Find the radius of the circle.

Part C**Find tangent**

Find the equation of the tangent to the circle at the point $P(8, 2)$. Give your answer in the form $ax + by + c = 0$, where a , b , and c are integers.

The following symbols may be useful: x , y

Part D**Find area**

The circle meets the y axis at Q and the tangent to the circle at P (as in part C) meets the y axis at R . Find the area of triangle PQR .

Used with permission from UCLES, A level, June 2015, Paper 4721, Question 10.

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Modulus 1i

Subject & topics: Maths Stage & difficulty: A Level P2

Part A

Transformations

Which transformations are needed to transform the graph of $y = |x|$ to the graph of $y = |2(x + 3)|$?

Choose two transformations from the list below and put them into order.

Available items

A stretch parallel to the y -axis with scale factor 2.

A stretch parallel to the x -axis with scale factor $\frac{1}{2}$.

A translation by 3 units in the negative x -direction.

A stretch parallel to the x -axis with scale factor 2.

A stretch parallel to the y -axis with scale factor 3.

A translation by 6 units in the negative x -direction.

A translation by 3 units in the positive x -direction.

Part B

Inequality

Solve the inequality $|x| > |2(x + 3)|$.

Construct your answer from the items below.

Items:

<

>

x

$< x <$

$\leq x \leq$

$< x \text{ or } x <$

$\leq x \text{ or } x \leq$

$\leq$

$\geq$

-7

-6

-5

-4

-3

-2

-1

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Functions: Reciprocal Trig 3i

Subject & topics: Maths **Stage & difficulty:** A Level P2

It is given that θ is the acute angle such that $\sec \theta \sin \theta = 36 \cot \theta$.

Part A

Value of $\tan \theta$

Find $\tan \theta$.

The following symbols may be useful: `cos()`, `sin()`, `tan()`, `theta`

Part B

Value of $\tan \left(\theta - \frac{\pi}{4} \right)$

Hence, using an appropriate formula, find the exact value of $\tan \left(\theta - \frac{\pi}{4} \right)$.

The following symbols may be useful: `pi`, `theta`

Part C

Value of $\tan (2\theta)$

Using an appropriate formula, find the exact value of $\tan (2\theta)$.

The following symbols may be useful: `theta`

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Trigonometry: Double Angles 2i

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Proof

Simplify $\sin(2\theta)(\tan\theta + \cot\theta)$ as far as possible.

The following symbols may be useful: theta

Part B

Exact Value

Hence find the exact value of $\tan\frac{\pi}{12} + \tan\frac{\pi}{8} + \cot\frac{\pi}{12} + \cot\frac{\pi}{8}$.

Part C

Solve

Using your answer to part A, solve the equation $\sin(4\theta)(\tan\theta + \cot\theta) = 1$ for $0 < \theta < \frac{\pi}{2}$, to three significant figures, giving your answer in **radians**.

Part D

$$(1 - \cos(2\theta))^2 \left(\tan \frac{\theta}{2} + \cot \frac{\theta}{2} \right)^3$$

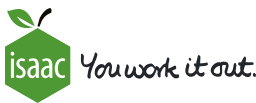
Using you answer to part A, express $(1 - \cos(2\theta))^2 \left(\tan \frac{\theta}{2} + \cot \frac{\theta}{2} \right)^3$ in terms of $\sin \theta$.

The following symbols may be useful: theta

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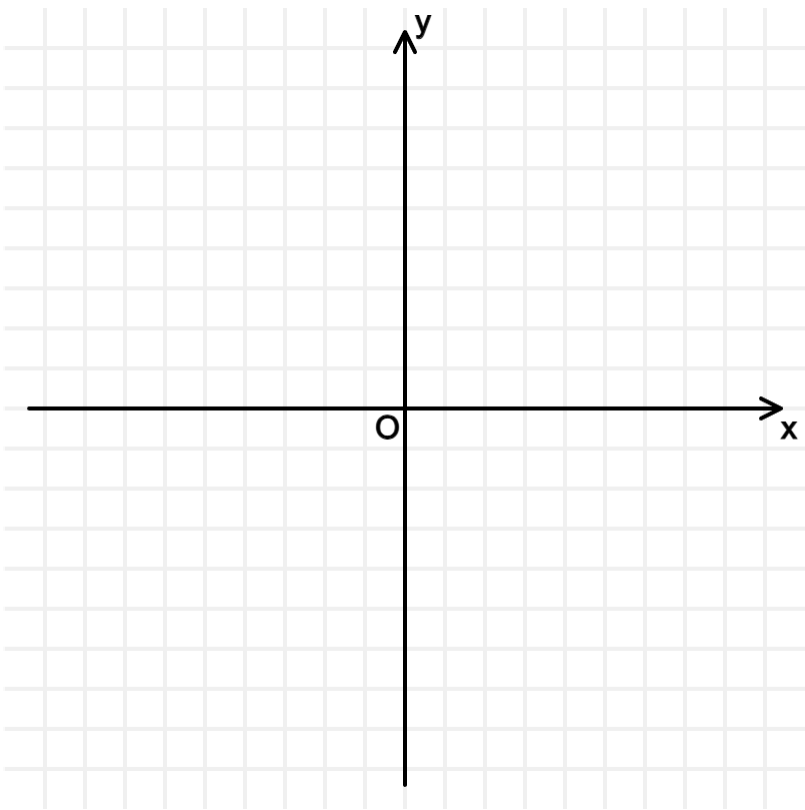
Sketching Inverse Trigonometric Functions 2

Subject & topics: Maths | Functions | Graph Sketching **Stage & difficulty:** A Level P2

Part A

Sketch $\arccos (x - 1) - 1$

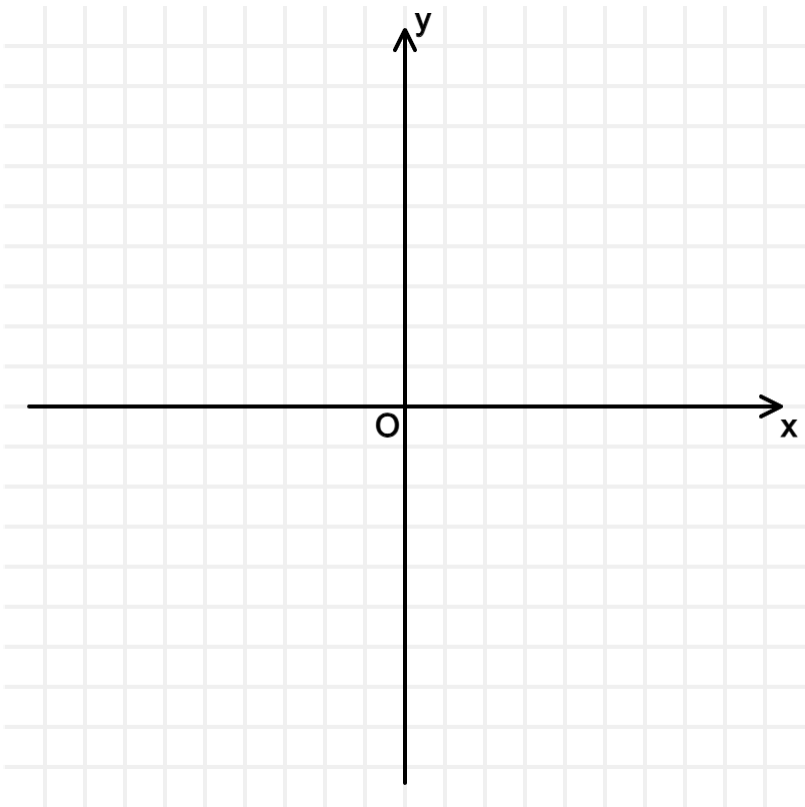
Sketch the graph of $y = \arccos (x - 1) - 1$.



Part B

Sketch $\frac{\pi}{2} - 2 \arctan x$

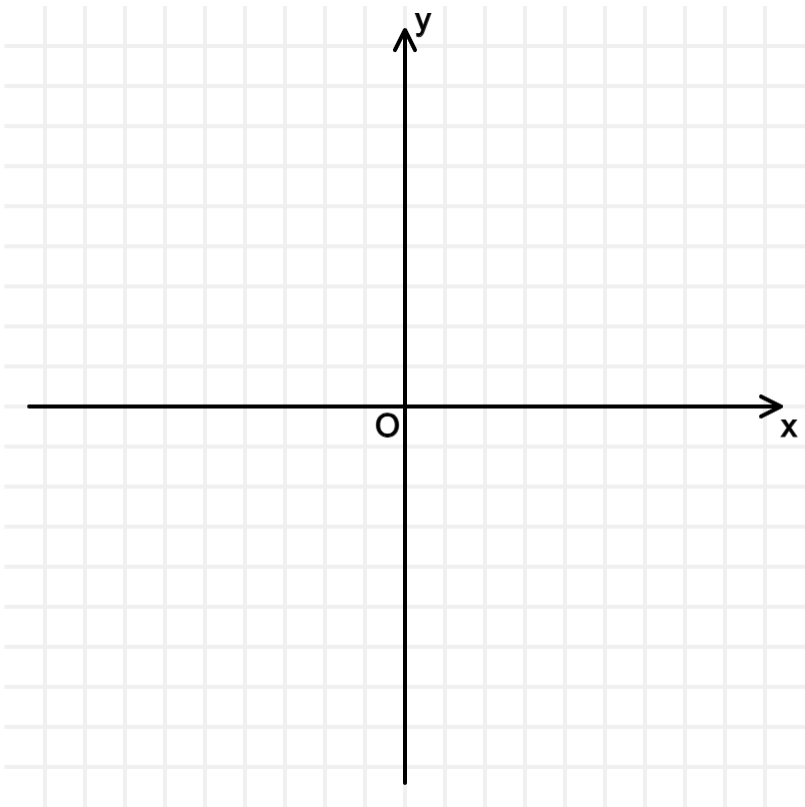
Sketch the graph of $y = \frac{\pi}{2} - 2 \arctan x$.



Part C

Sketch $\arcsin (2x - 1) - \frac{\pi}{2}$

Sketch the graph of $y = \arcsin (2x - 1) - \frac{\pi}{2}$.



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